

Liquid Pipeline Hydraulics

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Preface

This book presents liquid pipeline hydraulics as it applies to transportation of liquids through pipelines in a single-phase steady state environment. It serves as a practical handbook for engineers, technicians, and others involved in design and operation of pipelines transporting liquids. Existing books on the subject are mathematically rigorous and theoretical but lack practical applications. Using this book, engineers can better understand and apply the principles of hydraulics to their daily work in the pipeline industry without resorting to complicated formulas and theorems. Numerous examples from my experience are included to illustrate application of pipeline hydraulics.

The application of hydraulics to liquid pipelines involves an understanding of various properties of liquids, the concept of pressure, friction, and calculation of the energy required to transport liquid from point A to point B through a pipeline. You will not find rigorous mathematical derivation of formulas in this book. The formulas necessary for calculations are presented and described without using calculus or complex mathematical methods. The reader interested in how the formulas and equations are derived should refer to the books and other publications in the [References](#) section.

This book covers liquid properties that affect flow through pipelines, calculation of pressure drop due to friction, horsepower, and the number of pump stations required for transporting a liquid through a pipeline. Among the topics considered are the basic equations necessary for pipeline design, commonly used formulas to calculate frictional pressure drop and

necessary horsepower, the feasibility of improving an existing pipeline's performance using drag-reduction additives, and power optimization studies. The use of pumps and valves in pipelines is addressed along with modifications necessary to improve pipeline throughput. Economic analysis and transportation tariff calculations are also included.

The book can be used for the analysis of liquid pipeline gathering systems, plant or terminal piping, and long-distance trunk lines. The primary audience is engineers and technicians working in the petroleum, water, and process industries. It could also be used as a text for a college-level course in liquid pipeline hydraulics.

E.Shashi Menon

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